

The first Asian to win a Nobel Prize was an Indian. He was Rabindranath Tagore, the great poet. He looked at the whole world as one village. The great idea of a genius son of India is, in fact, on the verge of being achieved through the Internet.

TCP/IP and UDP/IP protocol suites

A protocol defines a set of rules for the functional operation of any computing system. As part of the process, a protocol resolves the problems of incompatibility and the overlaps existing in different systems. A protocol is also followed in offices. For example, a personal secretary follows a prescribed functional protocol to work with the boss. And, an attendant follows a different protocol so that there is no clash in running the office.

In 1984, International Standards Organization (ISO) of Europe developed an open systems interconnection (OSI) model for communication in networks. OSI is a layered reference framework of standards for communication among different parts of a network and/or among networks made of different equipment and applications by different vendors.

Incompatibilities are resolved by the application of reference model OSI. Over the years, OSI has been accepted as the basic and main architectural model for inter-computing, networking and inter-networking communications. As of today, most networks are based on OSI model. Seven layers of OSI reference model divide communication into seven peer processes, which are independent to each other.

The independent nature of implementing functions in each layer makes the design flexible and modular. The function of a layer may be set without affecting the functions of the other layers. For implementing the function of each layer, a set of rules

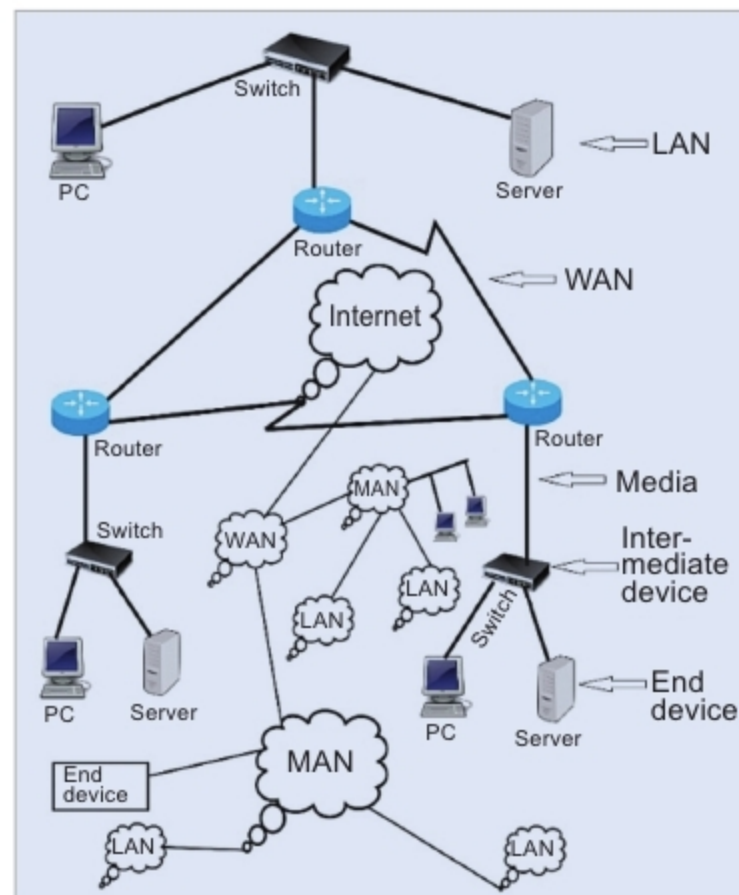


Figure 1: The Internet, a network of networks

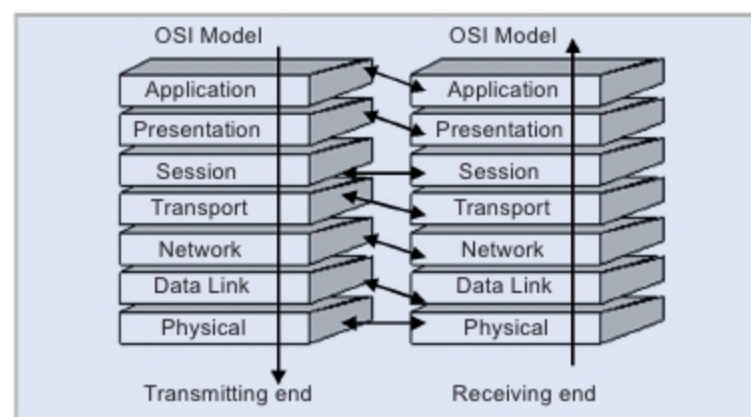


Figure 2: Illustration of OSI/ISO reference model

called protocol is defined. In OSI model, within each layer, one or more entities implement their functionality. Each entity interacts directly only with the layer immediately below it on the transmission end and provides facilities for use by the layer above it. Conceptually, at each layer at the transmitting end, a header is added with payload; and at the receiving end, a header is removed, as illustrated in Figure 2.

Functions of the seven layers of OSI model are:

- The physical layer defines the physical properties (for example, characteristics of signals and codes used for signals) of the various communication media used between a transmitter and a receiver. Physical properties include

electrical properties, type of cable used, type of BNC connector used, termination method, signal coding and so on. Cables may be coaxial, Ethernet, twisted pair or other. The layer therefore takes a decision on the method of transmission of bits as physical signals. The logical bit pipe received from the data link layer is converted into physical signals at the transmitter end. Physical signals received by the receiver are converted back into a logical pipe at the receiver end. This makes layer-to-layer functionality a peer process.

- The datalink layer does the logical frame of data bits transmitted/received over the physical layer. At the transmitting end, the layer generates a logical bit pipe with several overheads like address of source and destination, error check sum, length identification and so on, and at the receiving end, the bit pipe is decomposed (by removing the overheads after due processing/checking as the case may be) into received data bits.

Several protocols are used for the organisation of data bits at this layer. Examples of the protocols are

SDLC and HDLC. Data bits are put into a frame that includes headers of address (source and destination), control, check sum, and start and end of frame bytes.

LAN is functionally made up of datalink and physical layers. As LAN is a shared media network, the datalink layer for it has two functions: logical link control (LLC) sub-function and media access control (MAC) sub-function.

- The network layer performs routing and switching. It describes how data packets/frames traverse over various data links in the network between the source and destination. The layer defines the addressing and routing of the Internet. A node is made up of three layers: physical, datalink and network.
- The transport layer takes care of the quality and nature of data delivery. Quality of data delivery refers to issues like fast delivery versus